

Practical 4

AIM: To Implementation and Time analysis of factorial program using iterative and recursive method

Algorithm:

Algorithm for iterative method.

1. Start the program
2. Read the value fact = 1.
3. Repeat from 1 to n
4. multiplying fact by each number.
5. Display the factorial value.
6. Stop the program.

Algorithm for recursive method

1. Start the program.
2. Read the value of n.
3. If n is 0 (or) return 1.
4. otherwise, return n, fact (n-1).
5. Display the factorial value returned by recursive method.
6. Stop the program.

Program:

```
//iterative Method
#include <iostream> using
namespace std; class
Factorial
{ public:
```

```
int fact(int n) { int result
= 1; for (int i = 1; i <= n;
i++)
{
result = result * i;
} return
result;
} }; int
main() { int
n,result;
Factorial obj;
cout << "Enter the value to find its factorial: "; cin >>
n; result = obj.fact(n); cout << "Factorial of " << n << "
is " << result << endl; return 0;
}
//Recursive method
#include <iostream>
using namespace std;
class Factorial {
public:
int factr(int n) {
if (n == 0 || n == 1)
{ return
1; } else
{
int result = n * factr(n - 1); return
result;
}
} }; int
main() { int
n, result;
// Create an object of the class Factorial
obj;
cout << "Enter the value to find its factorial: ";
cin >> n; result = obj.factr(n); cout << "Factorial of "
<< n << " is " << result << endl; return 0; }
```

Output:

```
ENTER THE NUMBER TO GET FACTORIAL:5
ENTER 1 FOR ITERATIVE METHOD 2 FOR RECURSION METHOD :1
FACTORIAL IS: 120
```



```
ENTER THE NUMBER TO GET FACTORIAL:5
ENTER 1 FOR ITERATIVE METHOD 2 FOR RECURSION METHOD :2
120
```

Time Complexity:

time complexity

* loop executes 1 time.

* each iteration performs. are multiplication.

Best case: $O(n)$

Average case: $O(n)$

Worst case: $O(n)$

time complexity

* The recursive function is called once for each value.
n down to 1

Recursive relation

$$T(n) = T(n-1) + O(1)$$

Solving recurrence.

$$T(n) = O(n)$$

Best case: $O(1)$

when $n=0$ or $n=1$

Average case: $O(n)$

Worst case: $O(n)$

Conclusion:

The factorial was successfully implemented using iterative and recursive methods. Both methods have $O(n)$ time complexity. The iterative method uses $O(1)$ space, while the recursive method uses $O(n)$ space.